CSM
Spring 2026

EECS 16B
Worksheet 4

1. Complex Number Practice

(a) Take z_1 and z_2 to have the following rectangular coordinates form:

$$z_1 = 2 + 6j$$
$$z_2 = 3 - 5j.$$

What is the real part of $z_1 + z_2$?

Solution: 5

(b) What is the imaginary part of $z_1 + z_2$?

Solution: 1

(c) How do we visualize the polar representation of complex numbers, $z = Ae^{j\theta}$, in the complex plane?

Solution:

(d) Given $z = 7e^{j30^\circ}$, what is $|z|$?

Solution: 7

(e) What is $\angle z$?

Solution: 30°

(f) Given $z = 5e^{j90^\circ}$, what is the real part of z?

Solution: 0

(g) What is the imaginary part of z?

Solution: 5

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2. Introduction to Capacitors

Solution:
Description: Introduction to capacitors.
Prerequisite: Understanding of basic circuit components, voltage, current, resistors.
Mentors: See lecture notes for definition supplements

In this problem, we will introduce the fundamental circuit components.

(a) What is a capacitor?

Solution: Firstly, a capacitor is represented in this manner:



Physically speaking, it is two conductive plates separated by some (usually non-conductive) material. The parameters of interest here are:
A: the area of the surface of the plates that are facing each other.
ε: the permittivity of the material between the plates.
d: the distance between the two plates.

(b) What is capacitance?

Solution: Capacitance (C) is the measure of how much charge (Q) can be stored for some given amount of voltage (V). It is often written as:

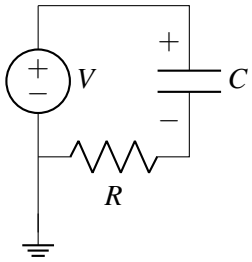
$$C = \frac{Q}{V} \text{ or } Q = CV$$

It can also be written as:

$$C = \frac{\varepsilon A}{d}$$

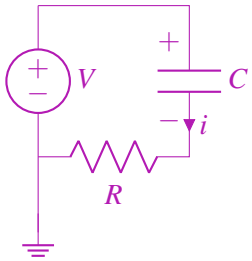
where ε is the "permittivity" of the material between the plates of the capacitor, A is the area of the surface of the plates facing each other, and d is the distance between the two plates.

(c) What happens to the following circuit (specifically with respect to the voltage across and current through the capacitor) when
a) the capacitor becomes fully charged?
b) after the capacitor is fully charged, we remove the voltage source and replace it with a short?

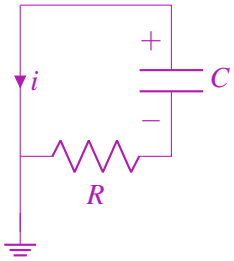


Solution: When the capacitor is fully charged, then there is no current flowing through the capacitor; $i = 0$.

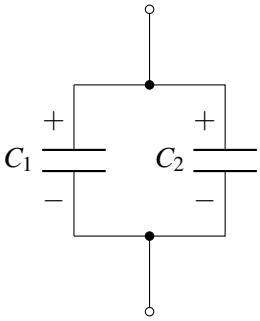
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If we replace the voltage source with a short, then the capacitor starts discharging because of the voltage difference across the resistor.



(d) Using physical properties of capacitors, derive the equation to find the equivalent capacitance for two capacitors in parallel, C_1 and C_2 in the figure below. Use this equation to find a relationship between the total charge of the two capacitors and the charge of each individual capacitor.



Solution: Recall the formula $C = \frac{\varepsilon A}{d}$. Let C_1 have an area of A_1, and let C_2 have an area of A_2. We can think of these capacitors as having a combined area of $A_1 + A_2$, so

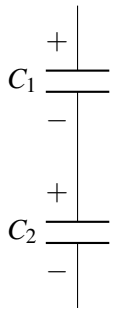
$$C_{eq} = \frac{\varepsilon(A_1 + A_2)}{d} = \frac{\varepsilon A_1}{d} + \frac{\varepsilon A_2}{d} = C_1 + C_2$$

Now, since $Q = CV$, the total charge $Q_{tot} = (C_1 + C_2)V$. We can leave V as is because the voltage is the same across both capacitors, since they share the same nodes. Then,

$$Q_{tot} = (C_1 + C_2)V = C_1V + C_2V = Q_1 + Q_2$$

(e) Using physical properties of capacitors, derive the equation for the equivalent capacitance of two capacitors in series.

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Solution: Let C_1 have a distance d_1 between its plates, and let C_2 have a distance d_2 between its plates. We can picture C_1 and C_2 as one capacitor with a combined distance $d_1 + d_2$. Then, assuming the areas for both capacitors are both A, we have $C_{eq} = \frac{\varepsilon A}{d_1+d_2}$. To make the math simpler, we invert both sides:

$$\frac{1}{C_{eq}} = \frac{d_1+d_2}{\varepsilon A} = \frac{d_1}{\varepsilon A} + \frac{d_2}{\varepsilon A} = \frac{1}{C_1} + \frac{1}{C_2}$$

Inverting it back, we get:

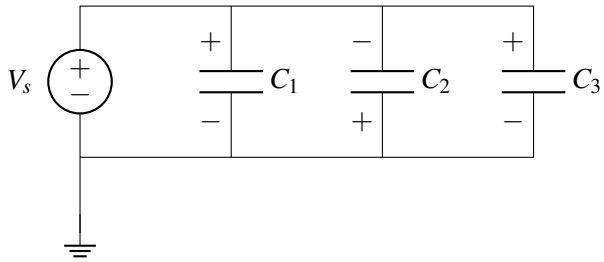
$$C_{eq} = \frac{1}{\frac{1}{C_1} + \frac{1}{C_2}} = \frac{C_1 C_2}{C_1 + C_2}$$

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3. Pitfall Problem

Solution: Prereq: Best placed right after the failure_cap_series_equivalence.tex
Description: Once students can do this, they can do every capacitor charge sharing problem (not necessarily those that include op amps too though)

(a) Consider the following circuit in ϕ_1. Assume that all capacitors are initially discharged. Find out the charge on each capacitor in this phase.



Solution:

$$Q_{C_1,\phi_1} = C_1 V_s$$
$$Q_{C_2,\phi_1} = -C_2 V_s$$
$$Q_{C_3,\phi_1} = C_3 V_s$$

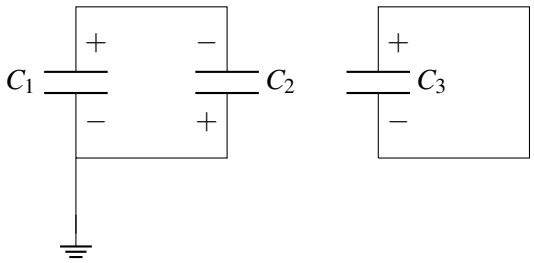
Note: What does it mean when we say that the charge on a capacitor is *negative*? It mesol that **on the positive plate of the capacitor, there is negative charge.** And since a capacitor has equal and opposite charge on each plate, the negative plate then has positive charge.

Solution: Make sure to stress the 'Note'. This is super important and is a misconception that students carry through the course.

$$Q = CV = C(V_+ - V_-)$$

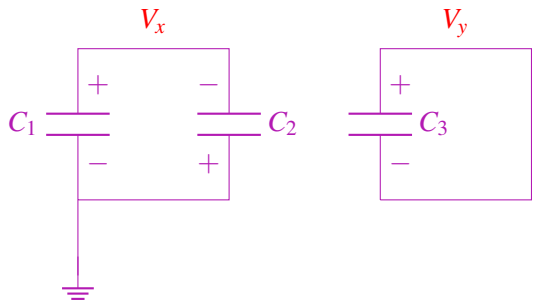
where V_+ and V_- are plates you arbitrarily mark as $+$ and $-$. It is obviously possible that you mark them incorrectly, and that there is actually negative charge on the positive plate and vice versa. If you use $Q = CV$ to calculate the charge on the capacitor, this always calculates the charge on the **positive** plate of the capacitor. If this quantity is negative, then there is negative charge on the plate you arbitrarily marked positive, and positive charge on the plate you arbitrarily marked positive. **This does not mean that you need to fix something in your circuit.**
This is fine. Just be consistent in the next phases.

(b) Assume that ϕ_1 has taken place, and that the capacitors are then moved to the following configuration in ϕ_2. Calculate the charge across each capacitor in ϕ_2.



Solution: Step I: Write the voltage drop across each capacitor.
If it cannot be determined, create variables till it can be determined.
Circuit redrawn with unknown voltages marked.

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$$V_{C_1,\phi_2} = V_x - 0 = V_x$$
$$V_{C_2,\phi_2} = 0 - V_x = -V_x$$
$$V_{C_3,\phi_2} = V_y - V_y = 0$$

Step 2: Write the charge on each capacitor.
Just use $Q = CV$ on the first step

$$Q_{C_1,\phi_2} = C_1 V_x$$
$$Q_{C_2,\phi_2} = -C_2 V_x$$
$$Q_{C_3,\phi_2} = 0$$

Step 3: Write the charge sharing equations on floating nodes.
Floating nodes are those where charges cannot escape or enter.
Node marked V_x is the only floating node

$$Q_{C_1,\phi_2} - Q_{C_2,\phi_2} = Q_{C_1,\phi_1} - Q_{C_2,\phi_1} \tag{1}$$

$$C_1 V_x - (-C_2 V_x) = C_1 V_s - (-C_2 V_s) \implies V_x = V_s$$

Final Step 4: Plug variable voltage into charge equations

$$Q_{C_1,\phi_2} = C_1 V_x = C_1 V_s$$
$$Q_{C_2,\phi_2} = -C_2 V_x = -C_2 V_s$$
$$Q_{C_3,\phi_2} = 0$$

Solution: Note: Make sure to explain the charge sharing equations properly.

Pitfall number 1: Students think that charge sharing equations always have positive signs on them. Something along the lines of $Q_{1,1} + Q_{2,1} = Q_{1,2} + Q_{2,2}$. Equation **??** shows that this is not the case. Explain clearly why there is a negative sign. This is because the positive plate of C_1 is connected to the negative plate of C_2. Charge is being redistributed by Q_{C_1} and $-Q_{C_2}$ in the two phases.

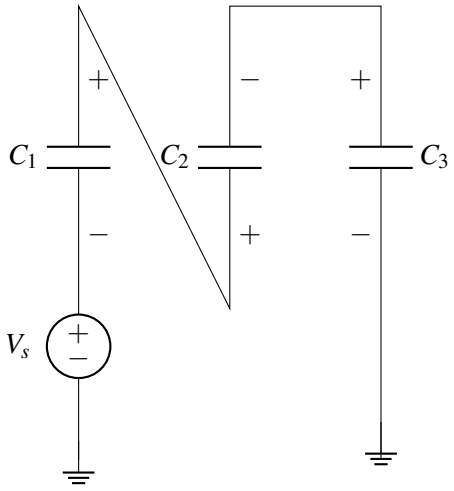
Pitfall number 2: Students think that charge sharing always happens between all capacitors. When you ask them for charge sharing equations, their gut reaction will be to say $Q_{C_1,\phi_1} + Q_{C_2,\phi_1} + Q_{C_3,\phi_1} = Q_{C_1,\phi_2} + Q_{C_2,\phi_2} + Q_{C_3,\phi_2}$. Show that them their gut reaction is wrong – because clearly C_3 is not involved

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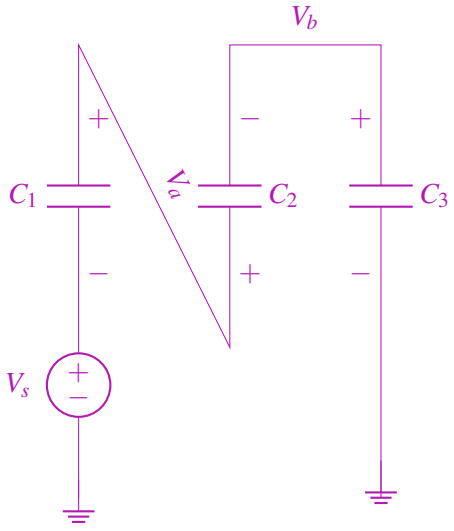
in charge sharing. Also show them that the plates affect the signs for charge sharing, it isn't always a plus sign as shown in Equation **??**.

Pitfall number 3: Many students don't realize that the voltage drop across C_3 is 0. They think that it will be the same as before and it will hold the same charge as before. This is not the case, as the math shows. Intuitively, this is not the case because + charges flow to recombine with the - charges resulting in 0 charge on the capacitor.

(c) Assume that ϕ_2 has taken place, and that the capacitors are then moved to the following configuration in ϕ_3. Calculate the charge across each capacitor in ϕ_3.



Solution: Step I: Write the voltage drop across each capacitor.
If it cannot be determined, create variables till it can be determined.
Circuit redrawn with unknown voltages marked.



$$V_{C_1,\phi_3} = V_a - V_s$$
$$V_{C_2,\phi_3} = V_a - V_b$$
$$V_{C_3,\phi_3} = V_b$$

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Step 2: Write the charge on each capacitor.
Just use $Q = CV$ on the first step

$$Q_{C_1,\phi_3} = C_1(V_a - V_s)$$
$$Q_{C_2,\phi_3} = C_2(V_a - V_b)$$
$$Q_{C_3,\phi_3} = C_3(V_b)$$

Step 3: Write the charge sharing equations on floating nodes.
Floating nodes are those where charges cannot escape or enter.
Nodes marked V_a and V_b are the only floating nodes

$$Q_{C_1,\phi_3} + Q_{C_2,\phi_3} = Q_{C_1,\phi_2} + Q_{C_2,\phi_2} \tag{2}$$

$$-Q_{C_2,\phi_3} + Q_{C_3,\phi_3} = -Q_{C_2,\phi_2} + Q_{C_3,\phi_2} \tag{3}$$

Plugging in values to Equation **??**

$$C_1(V_a - V_s) + C_2(V_a - V_b) = C_1V_s + (-C_2V_s)$$

Plugging in values to Equation **??**

$$-C_2(V_a - V_b) + C_3V_b = -(-C_2V_s) + 0$$

Final Step 4: Plug variable voltage into charge equations
We now have two equations in two variables (V_a, V_b), and so we can solve for them. After that, we can plug those into Step 2, and find the charges on each capacitor. There is no need to do this step as the expressions aren't very neat.

Solution: **Pitfall number 4**: Make sure to explain equations **??** and **??** very carefully. Students get confused specially on **??** because in ϕ_2, the positive plate of C_1 is connected to the negative plate of C_2. In ϕ_3, the positive plates of both are connected to each other. The thing is that we only need to look at the current phase ϕ_3's configuration to write the charge sharing equations. Previous configs don't matter. One way to explain this is by talking about phases as a 2 step process – disconnection and connection. First disconnect all capacitors, and all positive charges stay where they are and negative stay where they are. Now, if we connect the positive plate of two capacitors and if originally the positive and negative plates of two separate capacitors were connected earlier, the previous config doesn't matter. It is the two positive plate's charges that will need to stay conserved between phases. This is why the signs on **??** will be the same on both the left hand side of the equation and the right hand side, and you get the signs by simply looking at ϕ_3.

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